JV cover vs the matched negation-map rho: frozen re-scoring (2026-11-01)

Scope. The `jv_cover` genus-3 weak class only (never prime order), lower-bound comparison, no statement about prime-order or deployed curves. Rules are those of `PROTOCOL.md` (hashes in `results/FROZEN.sha`, frozen before any new run). The cover side is the 9 existing records; **no new cover runs**. Nothing here enters the leaderboard or `benchmarks/`.

Labels: **measured** = read from a run record; **derived** = computed from measured values by the Rust tool in `fit/`; **model** = extrapolation.

Files: `scaling.svg` / `scaling.png`, `REPORT.pdf`, `table.md`, `fit.txt` / `fit.json` (every statistic), `walls.txt`, raw records in `results/` (`new/` 108 new runs, `inputs/` copies of the 9 cover records, 18 old runs, 64 red-team runs, `replay/`), `run.sh`, `fit/`.

Headline (derived)

Against the **negation-map rho on the same instances** (frozen primary: estimator (b), 126 runs pooled over the three primes; homogeneity across p was not rejected), the cover costs **more** than rho in group additions at p = 1301 (S/rho 1.405) and p = 1451 (1.160, lower end 1.020, marginal), and is **not resolved** at p = 1601 (0.870 [0.697, 1.183]). In textbook F_p multiplications the cover costs more than rho at every prime (3.38, 2.79, 2.09). Distance remaining (the factor the cover must improve): 1.40 / 1.16 / none (not resolved, point 1.15x ahead) in U1, 3.38 / 2.79 / 2.09 in U2. This confirms the review's conclusion with a frozen design: the earlier "below rho at 1451 and 1601" claim holds only against the no-negation `rhodp`.

Measured improvement is the success criterion; this study measures the gap, it does not gate on it. The gap shrinks with p at about p^-2.1 (the crossover study's slope, here only used as a model).

Results

Result per prime, primary estimator (b) pooled new primes (frozen choice: homogeneity holds)

S/rho = geometric mean over the 3 cover seeds of S, divided by the reference mean; 95% percentile bootstrap, 20000 resamples of seeds and reference runs jointly (derived). Distance remaining = the factor by which the cover must improve to reach the matched rho (S/rho itself when above 1; the cover is ahead by 1/(S/rho) otherwise).

unit	p	cover geo-mean S (measured)	reference mean (derived)	S/rho [95% boot]	verdict	distance remaining
U1 group	1301	1.310	0.9322	**1.405**	above	x1.40 [1.21,

unit	p	cover geo-mean S (measured)	reference mean (derived)	S/rho [95% boot]	verdict	distance remaining
additions				[1.206, 1.629]		1.63]
U1 group additions	1451	1.081	0.9322	**1.160** [1.020, 1.338]	above	x1.16 [1.02, 1.34]
U1 group additions	1601	0.811	0.9322	**0.870** [0.697, 1.183]	not resolved	none: ahead by x1.15 [0.85, 1.44]
U2 textbook F_p mults	1301	1.310	0.3879	**3.376** [2.899, 3.915]	above	x3.38 [2.90, 3.92]
U2 textbook F_p mults	1451	1.081	0.3879	**2.787** [2.452, 3.215]	above	x2.79 [2.45, 3.21]
U2 textbook F_p mults	1601	0.811	0.3879	**2.090** [1.674, 2.842]	above	x2.09 [1.67, 2.84]

All three reference estimators (w = 0)

unit	estimator	n	ref mean	p=1301	p=1451	p=1601
U1 group additions	(a) per-prime	42/42/42	0.9203/1.0095/0.8669	1.423 [1.183, 1.716] above	1.071 [0.906, 1.286] not resolved	0.935 [0.717, 1.285] not resolved
U1 group additions	(b) pooled new primes	126	0.9322	1.405 [1.206, 1.629] above	1.160 [1.020, 1.338] above	0.870 [0.697, 1.183] not resolved
U1 group additions	(c) pooled + p=251	190	0.9111	1.437 [1.239, 1.657] above	1.187 [1.049, 1.363] above	0.890 [0.716, 1.213] not resolved
U2 textbook F_p mults	(a) per-prime	42/42/42	0.3830/0.4201/0.3607	3.420 [2.843, 4.123] above	2.573 [2.178, 3.090] above	2.248 [1.723, 3.087] above
U2 textbook F_p mults	(b) pooled new primes	126	0.3879	3.376 [2.899, 3.915] above	2.787 [2.452, 3.215] above	2.090 [1.674, 2.842] above
U2 textbook	(c) pooled +	190	0.3791	3.455	2.852	2.139

unit	estimator	n	ref mean	p=1301	p=1451	p=1601
F_p mults	p=251			[2.978, 3.982] above	[2.522, 3.277] above	[1.721, 2.915] above

Section 5a band (primary estimator; native operations priced at w group-addition equivalents each, on both arms)

Cover native count = `f4_native_ops` (emitted, a LOWER BOUND: F_p additions, pair selection, memory traffic not emitted). Reference native count = canon_cmp + hash + dp_check + scalar_ops + table_ops + cycle_cmp (emitted by the walk; `fp_adds` listed separately below, not priced in the band).

unit	p	w=0	w=0.1	w=1	same verdict at all w	cover native/sqrt(l) (lower bound)	rho native/sqrt(l)
U1 group additions	1301	1.405 [1.206, 1.629] above	1.467 [1.260, 1.702] above	1.498 [1.286, 1.738] above	yes (above)	23.94	15.81
U1 group additions	1451	1.160 [1.020, 1.338] above	1.211 [1.065, 1.397] above	1.236 [1.088, 1.426] above	yes (above)	19.71	15.81
U1 group additions	1601	0.870 [0.697, 1.183] not resolved	0.908 [0.727, 1.235] not resolved	0.927 [0.742, 1.261] not resolved	yes (not resolved)	15.12	15.81
U2 textbook F_p mults	1301	3.376 [2.899, 3.915] above	1.873 [1.608, 2.172] above	1.549 [1.330, 1.796] above	yes (above)	23.94	15.81
U2 textbook F_p mults	1451	2.787 [2.452, 3.215] above	1.545 [1.360, 1.783] above	1.278 [1.124, 1.474] above	yes (above)	19.71	15.81
U2 textbook F_p mults	1601	2.090 [1.674, 2.842] above	1.159 [0.928, 1.576] not resolved	0.958 [0.767, 1.304] not resolved	NO	15.12	15.81

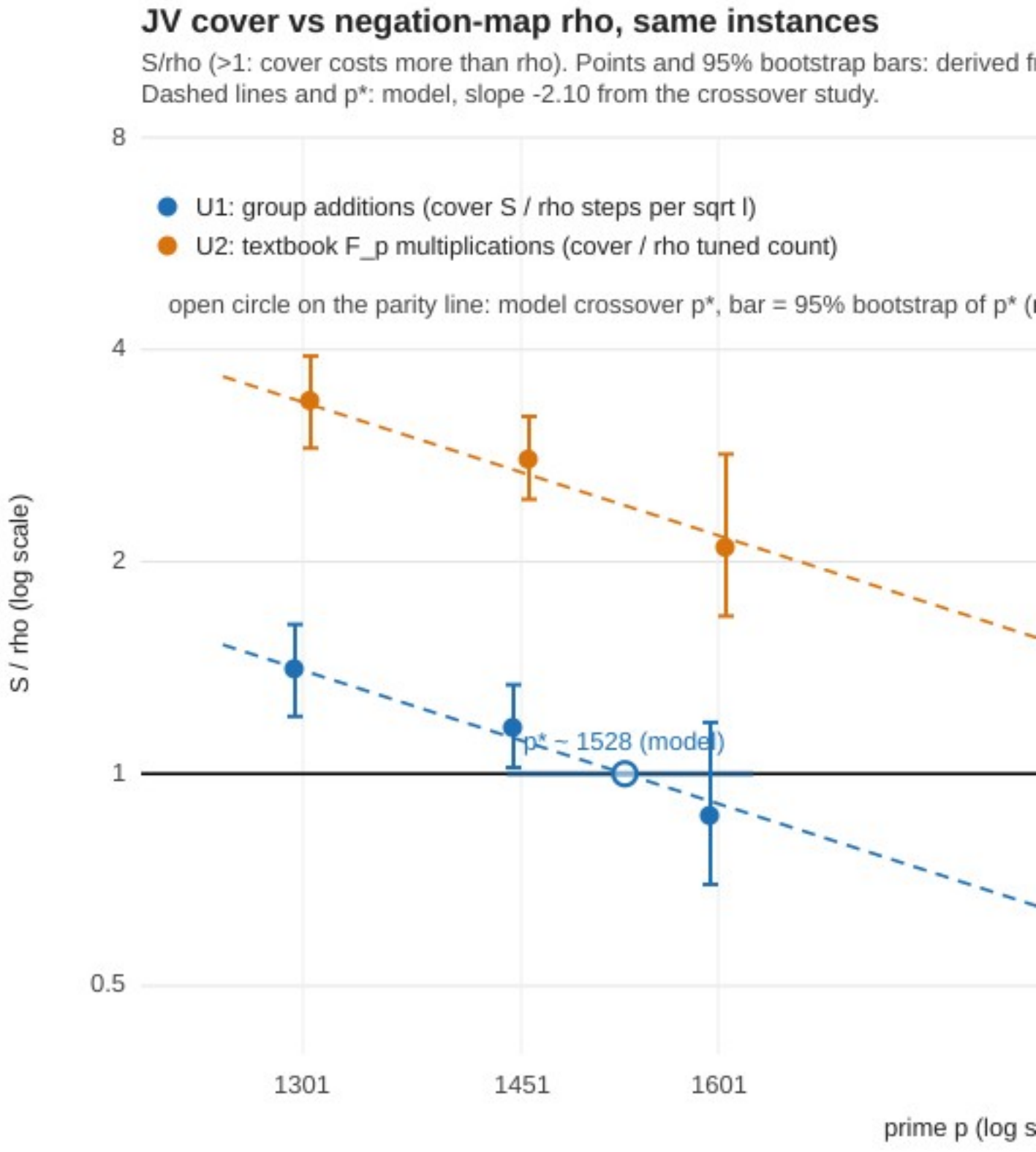
Unpriced counters listed (section 5a): reference `fp_adds` 42.1 per sqrt(l) (about 45.1 per step), emitted; cover F_p additions: NOT emitted (cover is a lower bound); cover residual-table and factor-base bytes: not operations, not charged (5a, 2026-11-01). Because the cover count is a lower bound, an "above" verdict at w = 0 can only widen under complete counts; a "below" or

"not resolved" verdict could flip.

Extrapolated fair crossover p* (model)

Model: S/rho(p) = G (p/p_i)^-2.10 through the three measured points (slope from the crossover study, not re-fitted here), p* where S/rho = 1. Interval: 2.5/97.5 percentile of the same bootstrap draws (no slope uncertainty). Slope sensitivity: -1.99 / -2.21 (the crossover study's 95% t interval).

unit	estimator	p* (model) [95% boot]	p* at slope -1.99	p* at slope -2.21	below the u64 limit p <= 2039?
U1 group additions	(a) per-prime	1529 [1443, 1633]	1534	1525	yes (whole interval)
U1 group additions	(b) pooled new primes	1528 [1441, 1629]	1533	1524	yes (whole interval)
U1 group additions	(c) pooled + p=251	1545 [1459, 1643]	1551	1540	yes (whole interval)
U2 textbook F_p mults	(a) per-prime	2322 [2191, 2479]	2384	2268	no: beyond the limit
U2 textbook F_p mults	(b) pooled new primes	2320 [2187, 2474]	2381	2266	no: beyond the limit
U2 textbook F_p mults	(c) pooled + p=251	2346 [2216, 2494]	2409	2290	no: beyond the limit


JV cover vs negation-map rho, same instances
S/rho (>1: cover costs more than rho). Points and 95% bootstrap bars: derived f
Dashed lines and p*: model, slope -2.10 from the crossover study.
U1: group additions (cover S / rho steps per sqrt l)
U2: textbook F_p multiplications (cover / rho tuned count)
open circle on the parity line: model crossover p*, bar = 95% bootstrap of p* (
S / rho (log scale)
8
4
2
1
0.5
p* ~ 1528 (model)
1301
1451
1601
prime p (log s

The extrapolation uses the crossover study's slope -2.10 (model, not re-fitted); the U1 crossover (about 1528) lies inside the implementation's u64 range p <= 2039, the U2 crossover (about 2320) lies beyond it: a cover that beats the matched rho in textbook F_p multiplications could not be run by this implementation without lifting the u64 limit. The old post-hoc red-team estimates were p* of 1568-1596 (U1) and 2327-2419 (U2).

Reference, homogeneity and controls (measured / derived)

- **Reference runs:** 14 per instance (2 old + 12 new, run indices 2..13, all 12 rounds complete, none excluded), 42 per prime, 126 at the new primes, 190 with the red team's 64 at p = 251. `l` and `c_add_e` of every run equal the cover record's.
- **Reference means (steps/sqrt l):** 0.920 (1301), 1.010 (1451), 0.867 (1601); pooled (b) 0.932, (c) 0.909; theory 0.888. s_harness and s_tuned: see `fit.txt`; s_tuned/s_harness = 0.41612-0.41613 on every run; s_harness differs from steps/sqrt(l) by -0.017% on average (max |.| 0.022%), so U1 equals group additions up to doublings, setup and batch products.
- **Homogeneity across p (steps/sqrt l):** one-way ANOVA F = 1.16, permutation p = 0.314; Kruskal-Wallis H = 2.08, permutation p = 0.354 (chi-square p 0.354). Not rejected, so the frozen primary is (b). With p = 251 added: 0.396 / 0.251. Across the 9 instances: 0.718 / 0.565. Power is low; "holds" means "not rejected".
- **Positive control: passed.** z against theory: (b) +1.15, (c) +0.64 (per prime: +0.51, +1.75, -0.31); all within 3 se. The pooled negation-map rho is consistent with sqrt(pi/4)/sqrt(1-1/256).
- **Consistency control: passed.** The tool reproduces the crossover study's S/rhodp and S/rhostrong (n = 6) point estimates and interval ends exactly to 3 decimals (e.g. 1.349 [1.085, 1.699], 1.063 [0.819, 1.477], 0.910 [0.658, 1.339]), and the red team's R4, R5, R6, R7 point estimates to 3 decimals (R5 1.480 / 1.221 / 0.916; R7 3.540 / 2.922 / 2.192; pooled 82 runs 0.8851 +- 0.0537); their interval ends agree within 0.01 under a different RNG stream.
- **Replay: bit-identical.** Four existing runs (p=1301 s1 run0, p=1451 s1 run0, p=1601 s1 run0, p=1601 s3 run1) rerun with the new binary equal the old records in every field except `wall_ms` (`results/replay/REPLAY.txt`). The binary (sha256 `8f62beea...8863`, git 3dcc7f2c7a) differs from the study's (`4a1284e9...9752`); only 4 of the 18 old runs were replayed.

Statement of what holds

1. Frozen primary: S/rho (U1) is above 1 at 1301 and 1451, not resolved at 1601; (U2) above at all three. At 1451 the U1 verdict is marginal (lower end 1.020 under (b); (a) gives 1.071 [0.906, 1.286], not resolved; (c) 1.187 [1.049, 1.363] above). It depends on the pooled estimator the protocol froze.
2. 3 cover seeds per prime remain the limit (10 distinct seed multisets). More cover seeds would be needed for any statement at 1601.
3. Section 5a: U1 verdicts are the same at w = 0, 0.1, 1 in all three cells (above, above, not resolved). In U2 the order holds at 1301 and 1451 (above) but **not at 1601**: above at w =

0, not resolved at w = 0.1 and 1. The cover native count is a lower bound; an "above" is robust to missing counts, "not resolved" is not. F_p additions are unpriced on the cover (not emitted) and listed for rho (42 per sqrt l), so a complete band cannot be computed.

Deviations and notes (plain)

1. **Runtime above projection.** The protocol projected K = 12 at about 2.8 h; actual 12906 s = 3.58 h (mean run wall 473 s, against the 368 s projected from the old runs: 337 / 508 / 574 s at 1301 / 1451 / 1601 with four jobs on four cores). All 108 runs launched before the 3 h 36 min launch deadline and all completed (exit 0); the cut rule did not bind. Study wall from the first command, about 4.4 h, within the 5 h cap.
2. **Tool edited after the freeze (layout only).** `fit/src/main.rs` was changed after freezing, only in the SVG layout (subtitle, label and legend positions, top margin); no statistic changed. Current sha256 prefix `af4e1355...` (frozen: `8efe0e95...`). The edit was made with a short Python text substitution; no number was computed in Python.
3. **Not routed through isolab-eks / not logged in the repository's usual places.** The brief specified this 4-core box, so the runs were local; I did not touch `docs/` (session log), conductor, resume checkpoints or the leaderboard, as instructed.
4. **Reference mixes binaries and dp_bits.** The 18 old runs and 64 red-team runs come from the study's binary; the 108 new runs from the rebuilt binary (replay identical on 4 runs). The p = 251 runs use `--dp-bits 12`, the others 16; only estimator (c) uses them.
5. The old rhostrong n = 6 per-prime means (0.971, 1.017, 0.892) moved to 0.920, 1.010, 0.867 with 42 runs per prime, as expected from their sampling error (se 0.14-0.19 before).
6. The cover records were measured with `JV_LIN` default "fast"; the review showed "sparse" would count 1/0.949 times as many multiplications. Cover S is used as recorded.
7. The pooled reference is slightly above theory (+1.15 se), not significant; no flag is raised.

Reproduce

```
cargo build --release --example jv_cover --offline      # binary sha256 in
results/versions.txt
research/jv_cover_negrho_rescore_20261101/run.sh         # 3.6 h on 4 cores
cd research/jv_cover_negrho_rescore_20261101/fit && cargo build --release --
offline && ./target/release/negrho_rescore ../results ..
```